

Problem 4:
You need to check
it's accurate

Cross-check extracted data against original

Combine three methods:

1. **“Eyeball” tests:** manual checks for known risks, vulnerabilities and issues
2. **Analysis tests:** calculations such as totals and value counts to cross-check
3. **AI tests:** document comparison and repeat of above tests

Eyeball test checklist:

- ✓ **Blanks/merged cells in original: check values haven't shifted**
- ✓ **Multi-line cells: check for no truncation**
- ✓ **First/last rows on PDF pages: check values extracted**
- ✓ **Headers & labels: check extracted and match data type**
- ✓ **Totals: check they are extracted/omitted as needed**
- ✓ **Non-numeric data: (“–”, “<5”): extracted accurately**
- ✓ **Spot-checks: for individual values at random**

Analysis test checklist:

- ✓ Values count: actual numbers vs expected (e.g. 5 cols x 8 rows)
- ✓ Non-numeric data: (“–”, “<5”): impacts on totals/counts
- ✓ Row totals*: calculate from values & match to PDF totals
- ✓ Column totals*: calculate & match to PDF
- ✓ Pivot table*: construct & match to PDF
- ✓ Sequence/ID continuity: check for gaps using formulae

**If original PDF includes totals/uses pivot table/sequential ID numbers*

AI test checklist:

Upload both documents

- ✓ Redo the same checks from eyeball and analysis tests
- ✓ Ask to suggest other checks that could be performed

Notes on validation: Cumbria

For the CSV extract:

- Eyeball test:
 - Figures not shifted where cells are empty
 - Data not affected where text runs across multiple lines
 - Spot-check: figures match PDF
 - Last/first row on pages: last row missing: Violence with Injury
 - Manual cleaning: Violence with injury row manually entered
 - Check row and column totals match totals in PDF
- Analysis test:
 - Check row totals match extracted totals: TRUE (aside from missing row)
 - Check column totals match extracted totals: TRUE (once missing row entered)
 - Columns x rows test: 32 categories x 5 years = 160
 - 104 numbers + 56 blank cells = 160
- AI test: “Attached is a CSV and PDF file of data. The CSV should contain the data from the table in the PDF associated with Question 1b: totals by offence sub group and year. Compare the data in the two. Identify any differences in either individual numbers or missing rows of data”
 -  Values: No numerical discrepancies at all once the one label is normalised. [cumbriaCLEAN | Excel], [Cumbria | PDF]
 -  Cosmetic issues: (a) CSV header typo 023/24 → 2023/24; (b) minor spacing in one offence name (Hacking-Personal vs Hacking- Personal).
 - Manual cleaning: years entered for those rows

Check	Completed by	Notes/action taken
Blanks/merged cells in original: check values haven't shifted		
Multi-line cells: check for no truncation		
First/last rows on PDF pages: check values extracted		
Headers & labels: check extracted and match data type		
Totals: check they are extracted/omitted as needed		
Non-numeric data: ("–", "<5"): extracted accurately		
Spot-checks: for individual values at random		
Values count: actual numbers vs expected (e.g. 5 cols x 8 rows)		
Non-numeric data: ("–", "<5"): impacts on totals/counts		
Row totals*: calculate from values & match to PDF totals		
Column totals*: calculate & match to PDF		
Pivot table*: construct & match to PDF		
Sequence/ID continuity: check for gaps using formulae		
AI check: value matching		Template here
AI check: row matching		
AI check: total checks		

Prompt template pt1: checking (new chat*)

**Make sure memory is turned off between chats*

CONTEXT

Attached is a CSV and the source PDF.

Focus only on the table(s) specified below and perform all of the following checks.

Report findings by section.

Target table: [DESCRIBE LOCATION/LABELS/STRUCTURE]

Ignore any tables that [DESCRIBE OTHERS]

If you cannot read all pages of the PDF (e.g., due to encryption or rendering limits), explicitly state which pages were validated and confine all findings to those pages. Recommend next steps for validating the remaining pages (e.g., obtaining a non-encrypted copy or exporting page by page).

Conclude with a one-line verdict: Accurate – no discrepancies found across all checks, or Discrepancies found – followed by a brief summary of the most significant issues.

Prompt pt2: value matching

CHECKS

1. Value matching

Compare every individual cell value between the PDF table and the CSV.

Report any numerical discrepancies as: *PDF value → CSV value* with row label and column/year.

Identify every instance where the PDF shows non-numerical values such as `–` or `<5` alongside numerical values. For each, state how it appears in the CSV, and assess whether the representation affects totals (e.g., `<5` stored as text vs a numeric value).

Note cosmetic differences separately (header typos, spacing, em-dash vs zero, etc)

Prompt pt3: row matching

2. Row matching

Flag any rows or categories present in the PDF but missing from the CSV, or extra rows in the CSV not in the PDF.

Check for systematic gaps in the CSV relative to the PDF, such as one missing row per page or missing first/last rows at page breaks.

If the PDF contains sequential IDs or numbering, list any missing IDs, infer any pattern (e.g., every 59th row absent), and provide examples.

Prompt pt4: total checks

3. Total checks

Compute the totals for the CSV and check those exactly match totals in the PDF (calculate those if not printed)

Do this for both row totals, column totals, and any other totals

Report any mismatches with the relevant row label and year column(s) .

Prompt pt5: label checks

4. Label checks

Compare all category, group, or label strings between the PDF and CSV.

Identify mismatches caused by punctuation, whitespace, or character encoding differences (e.g., curly apostrophes vs straight, en-dash vs hyphen).

For each discrepancy, map the PDF label to its CSV variant, estimate how many rows are affected, and confirm whether the underlying numeric values are otherwise aligned.

Prompt pt6: count checks

5. Count checks

Check that the CSV has the expected number of values based on the number of rows and columns in the PDF, excluding grand totals (e.g. 5 columns $\times$ 8 rows = 40 values)

Count how many values are in each row in the CSV, and how many are in each row in the PDF. Identify any mismatches.

What AI checks picked up

1. **Manually entered data** that contained errors (e.g. autocomplete wrongly completed)
2. **Artefacts of cleaning** (e.g. dashes removed in categories, or categories split on the dash)